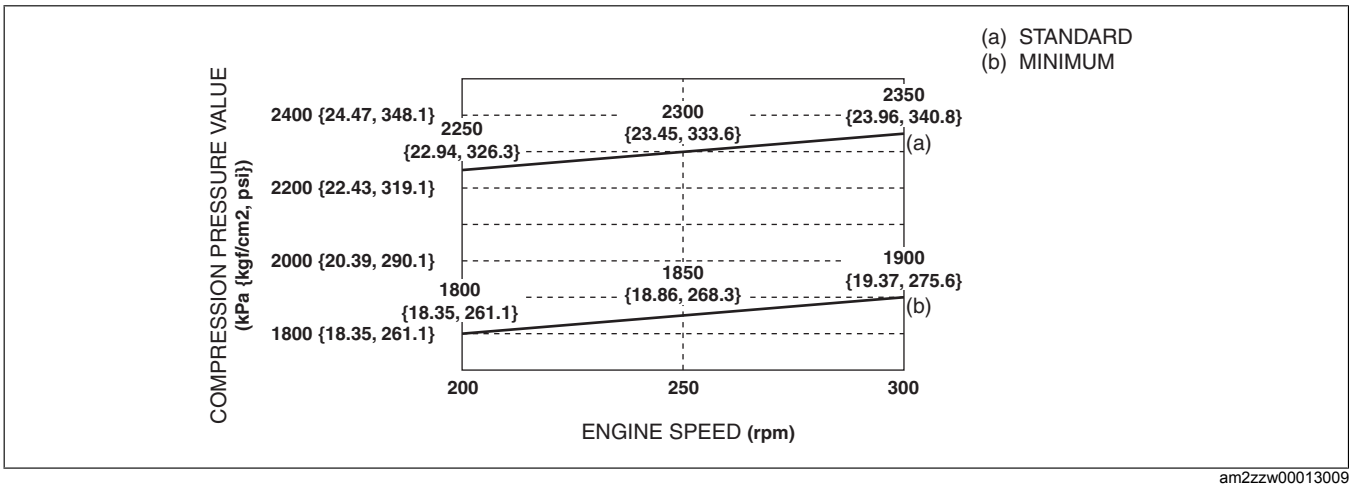
ENGINE TECHNICAL DATA [SKYACTIV-D 1.5]

id0150q1800100

Item	Specification
Difference between target idle speed and the engine speed	Within 50 rpm
Target idle speed (reference)	[MTX] Normal: 800 rpm DPF auto regeneration: 800 rpm Injector learning: 800 rpm [ATX] Normal: 775 rpm DPF auto regeneration: 800 rpm Injector learning: 800 rpm
Cylinder head bolt length L	L: 145.2—145.8 mm {5.717—5.740 in}
Cylinder head bolt length L maximum	L: 146.5 mm {5.768 in}
Front oil seal press on amount C	0—1.0 mm {0—0.039 in} from engine front cover
Rear oil seal press on amount C	0—0.5 mm {0—0.019 in} from cylinder block
Engine oil capacity (approx. quantity)	Oil replacement: 4.7 L {5.0 US qt, 4.1 Imp qt} Oil and oil filter replacement: 5.1 L {5.4 US qt, 4.5 Imp qt} Total (dry engine): 5.8 L {6.1 US qt, 5.1 Imp qt}
Oil pressure (reference value) [oil temperature: 80—90 °C {176—194 °F}, coolant temperature: 80—90 °C {176—194 °F}]	Lo: 110—175 kPa {1.13—1.78 kgf/cm^2, 16.0—25.3 psi} [1,500 rpm] Hi: 300—430 kPa {3.06—4.38 kgf/cm^2, 43.6—62.3 psi} [3,500 rpm]
Engine coolant capacity (approx. quantity)	MTX: 6.6 L {7.0 US qt, 5.8 Imp qt} ATX (Israel): 6.7 L {7.1 US qt, 5.9 Imp qt} ATX (Except Israel): 6.8 L {7.2 US qt, 6.0 Imp qt}
Water-cooled charge air cooler coolant capacity (approx. quantity)	2.0 L {2.1 US qt, 1.8 Imp qt}
Cooling system cap valve opening pressure	93.2—122.6 kPa {0.951—1.250 kgf/cm^2, 13.6—17.7 psi}
Water-cooled charge air cooler reserve tank cap valve opening pressure	24—36 kPa {0.25—0.36 kgf/cm^2, 3.5—5.2 psi}
Cooling fan motor standard current	16.0—20.0 A [12 V]
Glow plug resistance (specification)	Approx. 1.0 ohms [20 °C {68 °F}]
Turbocharger solenoid valve resistance (specification)	9.8—11.2 ohms [20 °C {68 °F}]
Fuel injector resistance (specification)	Approx. 0.59 ohms [20 °C {68 °F}]
Fuel pressure relief valve resistance (specification)	Approx. 0.47 ohms [20 °C {68 °F}]
Suction control valve resistance (specification)	Approx. 2.1 ohms [20 °C {68 °F}] If not within the specification, place a repair order with the DENSO, Ltd. services network.
Fuel warmer resistance (specification)	Approx. 0.85—1.28 ohms
HP-EGR control valve resistance (specification)	0.3—100 ohms [20 °C {68 °F}]
LP-EGR control valve resistance (specification)	0.3—100 ohms [20 °C {68 °F}]
Intake shutter valve resistance (specification)	0.3—100 ohms [20 °C {68 °F}]
Exhaust shutter valve resistance (specification)	0.3—100 ohms [20 °C {68 °F}]
Battery parasitic draw (When the ignition is off, all doors and the bonnet are closed.)	**Vehicles without i-ELOOP** • 40—65 mA **Vehicles with i-ELOOP** • 25—45 mA
Generator standard voltage [IG-ON] (Without i-ELOOP)	Terminal 1A: B+ Terminal 2A: Approx. 1 V or less Terminal 2B: Approx. 0 V
Generator standard voltage [Idle, 20 °C {68 °F}] (Without i-ELOOP)	Terminal 1A: 13—15 V Terminal 2A: Approx. 3—8 V Terminal 2B: Turn the electrical loads (such as headlights, blower motor, rear window defogger, brake lights) on and verify that the voltage reading increases.
Generator rotor resistance (between slip rings) [20 °C {68 °F}] (Without i-ELOOP)	1.8—2.1 ohms [20 °C {68 °F}]
Generator brush length (Without i-ELOOP)	Standard: 22.5 mm {0.886 in} Minimum: 5.0 mm {0.20 in}
Generator brush spring force (Without i-ELOOP)	Standard: 4.1—5.3 N {0.42—0.54 kgf, 1.0—1.1 lbf} Minimum: 1.7 N {0.17 kgf, 0.38 lbf}
Starter no-load test voltage	11 V

Item	Specification
Starter no-load test current	105 A or less

Compression pressure



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Standard: Refer to the graph.
Minimum: Refer to the graph.
Maximum difference between cylinders: 150 kPa {1.53 kgf/cm^2, 21.8 psi} [200 rpm]

Engine oil [U.K. specs.]

Recommended engine oil		Alternative engine oil
Mazda Original Oil Supra DPF 0W-30	**Mazda Original Oil Ultra DPF 5W-30**	**ACEA C3 0W-30, 5W-30**

Engine oil [Australian specs.]

Recommended engine oil
ACEA C1 or JASO DL-1 5W-30

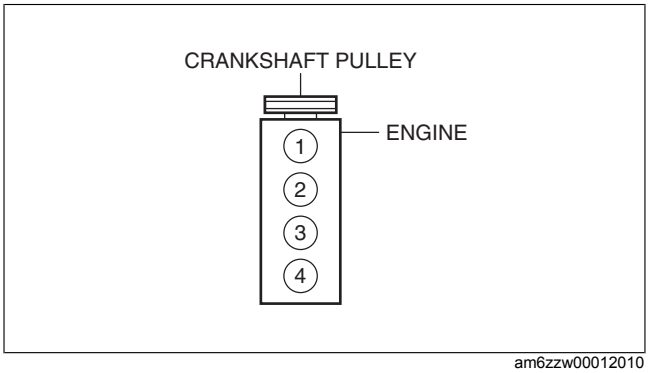
Engine oil [General (R.H.D.), Thailand specs.]

Recommended engine oil
ACEA C3 0W-30, 5W-30

Generator generated current (reference value) [Ambient temperature: 20 °C {68 °F}, Engine hot] (With i-stop, without i-ELOOP)

Engine speed (rpm)	Terminal 1A voltage (V)	Generator output current (A)
1,000	13	125
1,000	15	128
2,000	13	148
2,000	15	163

Cylinder number



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